

GENERIC DEGREES AND SPECIAL REPRESENTATIONS OF WEYL GROUPS

NOTES ON THE ATLAS SOFTWARE IMPLEMENTATION

CONTENTS

1. Introduction	1
2. Weyl Groups and Their Representations	1
2.1. Setup	1
2.2. The reflection representation	2
3. The Degree (b -Value)	2
3.1. Definition	2
3.2. Implementation in Atlas	2
4. Special Representations and Families	2
4.1. The Springer Correspondence	2
4.2. Special Orbits	3
4.3. Special Representations	3
4.4. Families	4
5. Generic Degree (a -Value)	4
5.1. Definition via Families	4
5.2. Definition via the Generic Hecke Algebra	4
5.3. Connection with Kazhdan–Lusztig Cells	5
5.4. Connection with Green Functions and Unipotent Characters	5
6. Computing Generic Degrees: Two Algorithms	5
6.1. Cell-Based Algorithm	5
6.2. Geck’s Inductive Algorithm	6
6.3. Mixed Strategy	7
7. Algorithm for the Map to the Special Representation	7
8. Display Functions in Atlas	8
9. Example: Type G_2	8
10. Summary	9
References	10

1. INTRODUCTION

Let $\mathbf{G}$ be a connected reductive algebraic group over an algebraically closed field, with Weyl group W . The irreducible complex representations of W play a central role in the representation theory of $\mathbf{G}$, both over finite fields (where they parametrize unipotent characters via Green functions) and in the study of primitive ideals in the universal enveloping algebra of the Lie algebra of $\mathbf{G}$.

Two numerical invariants are attached to each irreducible W -representation E :

- the *degree* (or *b-value*) $b(E)$, which is an integer measuring the first occurrence of E in the symmetric powers of the reflection representation;
- the *generic degree* (or *a-value*) $a(E)$, introduced by Lusztig, which controls the pole order of E in the generic Hecke algebra.

The inequality $a(E) \leq b(E)$ always holds, and equality $a(E) = b(E)$ characterizes a distinguished class of representations called *special representations*. These notions organize the irreducible W -representations into *families*, each containing exactly one special representation.

This document gives an account of these definitions and the algorithms implemented in the Atlas software to compute them. The primary references are Lusztig [4], Carter [1], and Geck [2].

2. WEYL GROUPS AND THEIR REPRESENTATIONS

2.1. Setup. Let $\mathbf{G}$ be a connected reductive group with maximal torus $\mathbf{T}$, root system $\Phi \subset X^*(\mathbf{T})$, and Weyl group $W = N_{\mathbf{G}}(\mathbf{T})/\mathbf{T}$. Fix a set of positive roots Φ^+ with simple roots $\alpha_1, \dots, \alpha_r$. For each $\alpha \in \Phi$ let $s_\alpha \in W$ denote the corresponding reflection.

The group W is a finite Coxeter group with generators $\{s_i = s_{\alpha_i} \mid 1 \leq i \leq r\}$. We write $\text{Irr}(W)$ for the set of isomorphism classes of irreducible complex representations of W .

2.2. The reflection representation. The *reflection representation* refl of W is the natural action of W on $\mathfrak{h}^* = X^*(\mathbf{T}) \otimes_{\mathbb{Z}} \mathbb{Q}$, which is faithful and irreducible for simple W .

3. THE DEGREE (b-VALUE)

3.1. Definition. The *module of coinvariants* of W is $\mathbb{C}[V]/(C[V]_+^W)$ where $V = \mathfrak{h}_{\mathbb{C}}^*$ and $(C[V]_+^W)$ is the ideal generated by W -invariant polynomials of positive degree. This module is graded and affords a graded W -module structure; its Poincaré series encodes the *fake degrees*.

More concretely, for each $E \in \text{Irr}(W)$ the *fake degree polynomial* $f_E(q) \in \mathbb{Z}[q]$ is defined by the multiplicity of E in the graded module of coinvariants:

$$f_E(q) = \sum_{k=0}^{|\Phi^+|} \langle E, \text{Sym}^k(\text{refl}) \rangle_W q^k.$$

(Here $\langle \cdot, \cdot \rangle_W$ denotes the inner product of characters of W .)

Definition 3.1. *The degree (or b -value) of $E \in \text{Irr}(W)$ is*

$$b(E) = \text{lowest degree of } q \text{ in } f_E(q) = \min\{k \geq 0 \mid \langle E, \text{Sym}^k(\text{refl}) \rangle_W \neq 0\}.$$

3.2. Implementation in Atlas. In the Atlas software (see `character_tables.at`), the character table of W is stored as a `CharacterTable` structure. The field `deg_spec` stores pairs $(b(E), \text{special}(E))$ for each irrep, computed at construction time. The degree $b(E)$ is found by searching for the smallest k such that $\langle E, \text{Sym}^k(\text{refl}) \rangle \neq 0$:

```
first(limit, (int k) bool: !=Wct.inner(char, Wct.sym_power_refl(k)))
```

The function `degree(ct)(i)` then returns the b -value of irrep i .

4. SPECIAL REPRESENTATIONS AND FAMILIES

4.1. The Springer Correspondence. The Springer correspondence [6] gives a bijection between the set of nilpotent orbits $\mathcal{O}$ of $\mathbf{G}$ (or more precisely, pairs $(\mathcal{O}, \phi)$ where ϕ is an irreducible character of the component group of the centralizer of a point of $\mathcal{O}$) and a subset of the set of irreducible W -representations.

For classical groups and in characteristic 0 or large characteristic, this gives a bijection between nilpotent orbits and a subset of $\text{Irr}(W)$. Explicitly:

- **Type A_{n-1} ($W = S_n$):** Irreps are indexed by partitions of n ; nilpotent orbits (Jordan types) are also indexed by partitions of n . The Springer correspondence sends each orbit directly to the corresponding irrep.
- **Type B_n / C_n :** Irreps are indexed by bipartitions (λ, μ) (ordered pairs of partitions with $|\lambda| + |\mu| = n$); nilpotent orbits are indexed by partitions of $2n$ (for C_n , those with each odd part occurring an even number of times; for B_n , each even part occurring an even number of times). The Springer map (via the core-quotient construction) sends each orbit to a bipartition.
- **Type D_n :** Similar to type B/C , with a complication for “split” classes and representations.

- **Exceptional types:** The Springer correspondence is given by explicit tables (see [1] and the Atlas files `springer_table_E6.at`, etc.).

4.2. Special Orbits. A nilpotent orbit $\mathcal{O}$ is *special* (in the sense of Lusztig) if it appears in the image of the Springer correspondence on the principal series, i.e., if $(\mathcal{O}, \text{triv})$ is in the image. Equivalently (for simply connected $\mathbf{G}$ in good characteristic), $\mathcal{O}$ is special if and only if the special closure of $\mathcal{O}$ equals $\mathcal{O}$ itself.

For type B_n/C_n , a partition (nilpotent orbit) is special if and only if its parts satisfy the appropriate parity conditions:

- Type C_n : each odd part occurs an even number of times.
- Type B_n : each even part occurs an even number of times.

4.3. Special Representations. Via the Springer correspondence, each special orbit $\mathcal{O}$ corresponds to an irreducible W -representation $E_{\mathcal{O}} = \text{Springer}(\mathcal{O})$. The representation $E_{\mathcal{O}}$ is called a *special representation* of W .

Equivalently, $E \in \text{Irr}(W)$ is special if and only if $a(E) = b(E)$, i.e., the generic degree equals the degree.

Definition 4.1. *An irreducible representation $E \in \text{Irr}(W)$ is special if $a(E) = b(E)$.*

In the Atlas software, the function `to_special` : $\text{Irr}(W) \rightarrow \text{Irr}(W)$ sends each irrep to its associated special representation (defined via the Springer map and Lusztig's $\mathbf{a}$ -function theory). The identity

$$\text{is_special}(E) \iff \text{to_special}(E) = E \iff \text{special}(E) = E$$

holds. For exceptional types, `to_special` is stored in explicit lookup tables (e.g., `to_special_E8` in `character_table_E8.at`):

```
set to_special_G2 = (int i) int:
    case i in 0, 5, 5, 3, 5, 5 else error("Wrong G2 irrep number") esac
```

For classical types, `to_special` is computed by the `make_special` function on bipartitions or symbols (in `combinatorics.at`).

4.4. Families.

Definition 4.2. *Two irreducible representations $E, E' \in \text{Irr}(W)$ are in the same family if $\text{to_special}(E) = \text{to_special}(E')$, i.e., if they share the same associated special representation.*

This defines a partition of $\text{Irr}(W)$ into families. Each family contains exactly one special representation. In the Atlas software (`families.at`):

```

set families(CharacterTable ct)=[[int]]:
let v = for i:ct.n_classes do [int]: [] od
in for j:ct.n_classes do v[ct.special(j)] #:= j od
; ...sort_special_first(ct,a)...

```

The function `families` returns a list of families, each sorted with the special representation first.

5. GENERIC DEGREE (a -VALUE)

5.1. Definition via Families.

Definition 5.1. *The generic degree (or a -value) of $E \in \text{Irr}(W)$ is*

$$a(E) = b(E_s)$$

where E_s is the unique special representation in the family of E .

Thus: $a(E) \leq b(E)$ with equality if and only if E is itself special. All representations in the same family share the same a -value.

In the Atlas software (`character_tables.at`, line 148):

```

set generic_degree (CharacterTable ct) = (int i)int: ct.degree(ct.special(i))

```

5.2. Definition via the Generic Hecke Algebra. Lusztig's original definition of the a -function is more algebraic. Let $\mathcal{H}$ be the generic Hecke algebra of W over $\mathbb{Z}[v, v^{-1}]$, with standard basis $\{T_w\}_{w \in W}$. Each irreducible W -representation E gives rise to an irreducible $\mathcal{H}$ -module E_v over the fraction field. The *generic degree* $a(E)$ is defined so that $v^{a(E)} \cdot c_{\text{leading}}$ is a unit in the $\mathbb{Z}[v]$ -form of the module, where c_{leading} comes from the structure constants in the Kazhdan–Lusztig basis $\{C_w\}$. More precisely, $a(E)$ is the minimum non-negative integer such that $v^{a(E)} \langle E_v, C_w \rangle$ lies in $\mathbb{Z}[v]$ for all $w \in W$ [4, Chapter 5].

5.3. Connection with Kazhdan–Lusztig Cells. A fundamental theorem of Lusztig [4] connects generic degrees with the Kazhdan–Lusztig cell theory:

Theorem 5.2 (Lusztig). (1) *Each two-sided Kazhdan–Lusztig cell $\mathbf{c}$ of W contains exactly one special representation E_s .*
 (2) *All irreps E appearing in the cell character of $\mathbf{c}$ satisfy $a(E) = b(E_s)$.*
 (3) *The special representation E_s in $\mathbf{c}$ is the unique irrep of minimum b -value among those appearing in the cell character of any left cell contained in $\mathbf{c}$.*

The correspondence $\mathbf{c} \leftrightarrow E_s$ gives a bijection between the set of two-sided cells of W and the set of special representations; thus the families of $\text{Irr}(W)$ are in bijection with two-sided cells.

5.4. Connection with Green Functions and Unipotent Characters. When $\mathbf{G} = \mathbf{G}(\mathbb{F}_q)$ is a finite group of Lie type, the generic degree $a(E)$ appears in the pole order of the generic degree polynomial of the unipotent character associated to E via the Deligne–Lusztig theory. The Green functions $Q_L^{\mathbf{G}}$ encode the Springer correspondence and the a -values control the leading behavior in q of matrix coefficients.

6. COMPUTING GENERIC DEGREES: TWO ALGORITHMS

The Atlas software implements two approaches to computing generic degrees.

6.1. Cell-Based Algorithm. The first algorithm computes generic degrees from Kazhdan–Lusztig cells (implemented in `generic_degrees.at`):

- (1) Compute the Kazhdan–Lusztig left cells of W (via the built-in `W_cells_of` function, which uses the KL polynomials of a block of the representation-theoretic category $\mathcal{O}$).
- (2) For each left cell C , compute the *cell character*: the character of W acting on the cell module, $\chi_C = \sum_{w \in C} (-1)^{\ell(w)} \chi_w$ (in appropriate sense).
- (3) Decompose χ_C into irreducibles: $\chi_C = \sum_E m_E(C) \cdot E$.
- (4) Identify the *special irreducible* in C : the unique irrep of smallest b -value among those with $m_E(C) \neq 0$:

```

set special_irreducible (CharacterTable ct, [int] multiplicities) = int
  let summands = smallest_degree_summands(ct, multiplicities)
in case #summands
  in error("zero character, no special")
  , summands[0]
  else error("non-unique special representation")
esac

```

- (5) The generic degree of every irrep E appearing in χ_C is $a(E) = b(E_s)$ where E_s is this special irreducible.

This method is efficient when the cells can be computed (which requires computing KL polynomials of a block), but may be slow or infeasible for very large rank groups (e.g., E_8) due to the size of the blocks.

6.2. Geck’s Inductive Algorithm. The second algorithm, due to Geck [2], computes the a -function inductively on the rank (implemented in `geck_generic.at`).

Setup. For a Weyl group W with root system Φ , let $\mathcal{L}$ denote the set of proper parabolic subgroups (Levi factors of proper parabolic subgroups), i.e., the Weyl groups $W_J = W(\Phi_J)$ for $J \subsetneq \{1, \dots, r\}$. The inductive hypothesis is that $a(E')$ is known for all proper subgroups W_J and all $E' \in \text{Irr}(W_J)$.

Key quantity. For $E \in \text{Irr}(W)$, define

$$a'(E) = \max\{a(E') \mid L \in \mathcal{L}, E' \in \text{Irr}(W_L), \langle \text{Ind}_{W_L}^W(E'), E \rangle \neq 0\},$$

i.e., $a'(E)$ is the maximum a -value over all proper Levi subgroup representations that appear in the induced representation. In Atlas (`geck_generic.at`):

```
set a_prime(RootDatum G, [CharacterTable] tables,
            int table_index, int character_index) = int:
max( for M in max_Levi_subgroups(G) do
    for char_M@t in ct_M.characters do
        if ind[character_index]>0 then [ct_M.generic_degree(t)] else [] fi
    od
od )
```

The ω_L correction. For $E \in \text{Irr}(W)$, define a correction term involving the character values at reflections. Following Geck [2, Definition 4.1]:

$$\omega_L(E) = \frac{1}{\dim E} \sum_{C_s} |C_s| \cdot \chi_E(s)$$

where the sum runs over conjugacy classes C_s of reflections $s \in W$, and χ_E denotes the character of E . (Here $|C_s|$ is the size of the class.)

The formula. Geck's formula is:

$$(1) \quad a(E) = \max(a'(E), a'(E \otimes \text{sgn}) - \omega_L(E))$$

where sgn is the sign character of W .

In Atlas (`geck_generic.at`):

```
let A = a_prime(G, tables, table_index, character_index)
, B = a_prime(G, tables, table_index,
              table.tensor_sign_index(character_index))
, omega_L = omega_L(table, character_index, ...)
then max = max(A, B - omega_L)
in table.set_generic_degree(character_index, d, true)
```

Inductive structure. The algorithm builds a list of character tables for all Levi subgroups (from small to large), filling in generic degrees inductively. The function `all_levi_tables` computes tables for all Levi subgroups, ordered so that proper Levis of any given group appear later in the list:

```

set set_generic_degrees_geck(RootDatum G) = CharacterTable:
  update_generic_degrees_geck(all_levi_tables(G))

```

6.3. Mixed Strategy. In practice, Atlas uses a combined strategy (`set_generic_degrees_3` in `geck_generic.at`):

- (1) First attempt to compute generic degrees using the cell algorithm for all Levi subgroups where cells are feasible.
- (2) Then use the sign symmetry: since $a(E \otimes \text{sgn}) = a(E) + \omega_L(E)$, knowing $a(E)$ immediately gives $a(E \otimes \text{sgn})$ (function `update_generic_degrees_using_sign`).
- (3) Finally, apply Geck's formula to any remaining irreps.

```

set set_generic_degrees_3 (RootDatum G) = [CharacterTable]:
( let tables = all_levi_tables_plus(G)
  in update_generic_degrees_using_sign(tables)
  ; update_generic_degrees_geck(tables)
  ; tables
)

```

7. ALGORITHM FOR THE MAP TO THE SPECIAL REPRESENTATION

Given $E \in \text{Irr}(W)$, a separate algorithm (in `special_rep.at`) computes the map $E \mapsto E_s$ (the special representation in the same family) without going through families. This relies on the following theorem:

Theorem 7.1 (Lusztig, [4]). (a) *If L is a Levi subgroup of G , and C_L is a left cell of W_L , then $\text{TGInd}(C_L)$ is a left cell of W .*
 (b) *If C is a left cell of W , so is $C \otimes \text{sgn}$.*
 (c) *If $W \neq 1$, then for any left cell C of W there is a proper Levi subgroup L and a left cell C_L of W_L such that either $\text{TGInd}(C_L) = C$, or $\text{TGInd}(C_L \otimes \text{sgn}) = C \otimes \text{sgn}$.*

Here TGInd denotes truncated (or j -)induction, which preserves cells and families. The algorithm to compute $E \mapsto E_s$ is:

- (1) If E is already special, return E .
- (2) If W is abelian, return the trivial representation.
- (3) Use part (c) of the theorem: find a proper Levi L and $E_L \in \text{Irr}(W_L)$ such that E appears in $j\text{-Ind}_{W_L}^W(E_L)$ with $a(E_L) = a(E)$. By induction, compute $(E_L)_s$. Then $E_s = j\text{-Ind}_{W_L}^W((E_L)_s)$.
- (4) If step 3 fails for E , replace E with $E \otimes \text{sgn}$, which is guaranteed to succeed; then apply the duality involution on special orbits to recover E_s .

For exceptional types, the map $E \mapsto E_s$ is given by explicit tables (e.g., in `character_table_G.at`, `character_table_E8.at`, etc.), while

for classical types it is computed via the combinatorics of bipartitions or symbols.

8. DISPLAY FUNCTIONS IN ATLAS

The Atlas software provides several convenience functions for inspecting these invariants:

- `show(ct)`: Displays a table of all irreducible representations with their dimensions, the index of their special representation (marked with `*` if self-special), the b -value `deg`, and the generic degree `gdeg`.
- `show_degrees(ct)`: Alternate display comparing $a(i)$ and $b(i)$, with stars marking special representations.
- `families(ct)`: Returns the partition of $\text{Irr}(W)$ into families, with each family sorted so the special representation appears first.
- `generic_degree(ct)(i)`: Returns $a(E_i)$ for the i -th irrep.

9. EXAMPLE: TYPE G_2

The Weyl group $W(G_2)$ has six irreducible representations, labeled (in the Atlas conventions) $0, 1, \dots, 5$:

i	Name	dim	special(i)	$a(E_i) = b(E_{\text{spec}})$
0	trivial	1	0 (*)	$b(0)$
1	short sign	1	5	$b(5)$
2	long sign	1	5	$b(5)$
3	full sign	1	3 (*)	$b(3)$
4	triangular reflection	2	5	$b(5)$
5	hexagonal reflection	2	5 (*)	$b(5)$

The `to_special_G2` function (from `character_table_G.at`) is:

```
set to_special_G2 = (int i) int:
    case i in 0, 5, 5, 3, 5, 5 else error("Wrong G2 irrep number") esac
```

encoding the map $0 \mapsto 0, 1 \mapsto 5, 2 \mapsto 5, 3 \mapsto 3, 4 \mapsto 5, 5 \mapsto 5$.

The special representations are those marked (*): 0 (trivial), 3 (full sign), and 5 (hexagonal reflection). The families are:

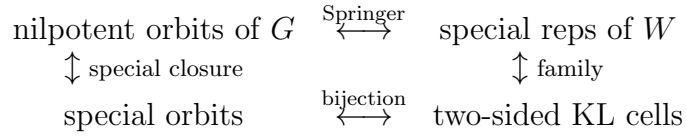
- $\{0\}$, with special rep 0;
- $\{3\}$, with special rep 3;
- $\{1, 2, 4, 5\}$, with special rep 5.

All four representations in the third family share the same generic degree $a = b(5)$.

Remark 9.1. *In type G_2 , the sign character splits into two distinct one-dimensional representations (short sign and long sign) reflecting the two root lengths. Both, together with the triangular reflection, lie in the family of the hexagonal reflection (the unique 2-dimensional special representation with the lower b -value), illustrating that non-singleton families can contain both one- and two-dimensional representations.*

10. SUMMARY

The following diagram summarizes the relationships:



For each $E \in \text{Irr}(W)$:

$$b(E) = \min\{k \mid \langle E, \text{Sym}^k(\text{refl}) \rangle \neq 0\} \quad (\text{degree, } b\text{-value, fake degree exponent})$$

$$E_s = \text{unique special rep in family of } E \quad (\text{via } \text{to_special})$$

$$a(E) = b(E_s) \quad (\text{generic degree, } a\text{-value})$$

The Atlas software computes $a(E)$ by either:

- the **cell algorithm**: computing KL cells and finding the minimum- b irrep in each cell; or
- **Geck's inductive formula** $a(E) = \max(a'(E), a'(E \otimes \text{sgn}) - \omega_L(E))$, proceeding inductively on Levi subgroups.

In practice a mixed strategy is used, applying the cell algorithm where feasible and the Geck formula for the remainder.

REFERENCES

- [1] R. W. Carter, *Finite Groups of Lie Type: Conjugacy Classes and Complex Characters*, Wiley Classics Library, John Wiley & Sons, 1985.
- [2] M. Geck, *Some applications of CHEVIE to the theory of algebraic groups*, Canad. J. Math. **61** (2009), 1239–1261. (Cf. also: M. Geck, *Some applications of Chevie to the theory of algebraic groups*, preprint.)
- [3] D. Kazhdan and G. Lusztig, *Representations of Coxeter groups and Hecke algebras*, Invent. Math. **53** (1979), 165–184.
- [4] G. Lusztig, *Characters of Reductive Groups over a Finite Field*, Annals of Mathematics Studies **107**, Princeton University Press, 1984.
- [5] G. Lusztig, *A class of irreducible representations of a Weyl group*, Indag. Math. **41** (1979), 323–335; *A class of irreducible representations of a Weyl group, II*, Indag. Math. **44** (1982), 219–226.
- [6] T. A. Springer, *A construction of representations of Weyl groups*, Invent. Math. **44** (1978), 279–293.